

413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 5

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RECALL

Recall from the previous lecture that we observe independent and identically distributed random variables

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p, \quad \mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2.$$

For the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i,$$

we showed

$$\mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}, \quad \text{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}.$$

Hence, averaging preserves the mean while fluctuations shrink as

$$\text{uncertainty in the sample mean} \sim \frac{1}{\sqrt{n}}.$$

1 Law of Large Numbers

We can make the concentration of $\bar{X}_n$ around μ precise using Chebyshev's inequality. Since

$$\mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n},$$

Chebyshev's inequality gives, for every $\varepsilon > 0$,

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

As $n \rightarrow \infty$,

$$\frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

Therefore,

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \rightarrow 0$$

for every $\varepsilon > 0$. We write this as

$$\bar{X}_n \xrightarrow{P} \mu.$$

This is the **Law of Large Numbers (LLN)**. In words:

As the number of independent observations increases, the sample mean $\bar{X}_n$ becomes increasingly concentrated around the population mean μ .

The LLN therefore tells us *where* the sample mean goes:

$$\boxed{\bar{X}_n \longrightarrow \mu.}$$

But it leaves another important question:

What is the shape of the fluctuations of $\bar{X}_n$ around μ ?

2 Central Limit Theorem

The Law of Large Numbers tells us that

$$\bar{X}_n \xrightarrow{P} \mu.$$

But how does $\bar{X}_n$ fluctuate around μ for large but finite n ? Recall that

$$\mathbb{E}[\bar{X}_n] = \mu, \quad \text{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}.$$

Thus, the natural scale of the fluctuations is $\sigma/\sqrt{n}$. We therefore define the **standardized sample mean**

$$\boxed{Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}.}$$

Notice that

$$\mathbb{E}[Z_n] = 0, \quad \text{Var}(Z_n) = 1.$$

The **Central Limit Theorem (CLT)** states that, for $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p$ with

$$\mathbb{E}[X_i] = \mu, \quad 0 < \text{Var}(X_i) = \sigma^2 < \infty,$$

we have

$$\boxed{\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).}$$

Equivalently, for sufficiently large n ,

$$\boxed{\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).}$$

Thus, the CLT gives the approximate **shape** of the sampling distribution:

Center:	μ ,
Width:	$\sigma/\sqrt{n}$,
Shape:	approximately Gaussian.

The LLN and CLT therefore answer two different questions:

LLN:	$\bar{X}_n \xrightarrow{P} \mu$	Where does the sample mean go?
CLT:	$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$	What is the shape of its fluctuations?

Importantly, the CLT does *not* say that the original distribution $p(x)$ is Gaussian. The population may be discrete, asymmetric, or bimodal; it is the **sample mean** that becomes approximately Gaussian for sufficiently large n .

3 Example: Monte Carlo Estimation of π

We draw n points (X_i, Y_i) independently and uniformly from the square $[-2, 2]^2$. For each sampled point, we define the indicator random variable

$$Z_i = \begin{cases} 1, & X_i^2 + Y_i^2 \leq 1, \\ 0, & X_i^2 + Y_i^2 > 1. \end{cases}$$

Thus, every randomly drawn point corresponds to a single **Bernoulli trial**: a point inside the unit circle is a success ($Z_i = 1$, shown in red), while a point outside the circle is a failure ($Z_i = 0$, shown in blue in the figure).

Since the sampling is uniform, the probability of success is determined entirely by the ratio of the two areas,

$$p = \mathbb{P}(Z_i = 1) = \frac{\text{area of unit circle}}{\text{area of square}} = \frac{\pi}{16}.$$

Consequently,

$$Z_i \sim \text{Bernoulli}\left(\frac{\pi}{16}\right).$$

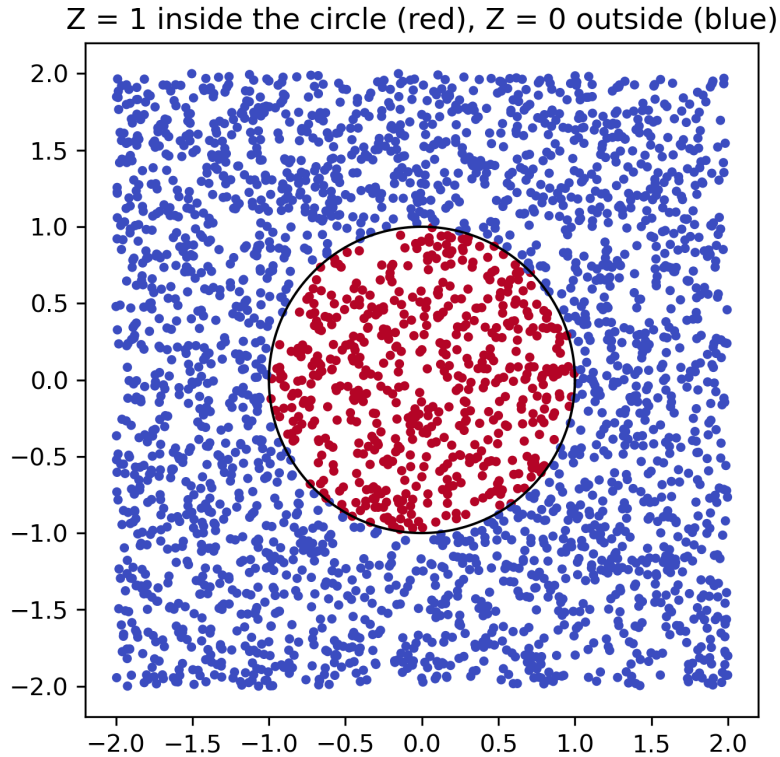


Figure 1: **Monte Carlo experiment for estimating π .** Points (X_i, Y_i) , $n = 3000$ are sampled independently and uniformly from $[-2, 2]^2$. A point is assigned $Z_i = 1$ if it lies inside the unit circle (red) and $Z_i = 0$ otherwise (blue). Since the ratio of the circle area to the square area is $\pi/16$, each draw is a Bernoulli trial with success probability $p = \pi/16$. The observed fraction of red points therefore provides an estimate of $\pi/16$.

Since, Z_i is a Bernoulli trial,

$$\mathbb{E}[Z_i] = p = \frac{\pi}{16}, \quad \text{Var}[Z_i] = p(1 - p) = \frac{\pi}{16}\left(1 - \frac{\pi}{16}\right),$$

We therefore repeat n independent Bernoulli trials, each with probability of success $p = \frac{\pi}{16}$ and variance $p(1-p)$. If we draw from this distribution n samples, and estimate the sample mean which is also a random variable,

$$\bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i,$$

is the fraction of points that fall inside the disk. From the properties of the sample mean, we know that the expectation of the mean and variance are given as,

$$\mathbb{E}[\bar{Z}_n] = \mu = p, \quad \text{Var}[\bar{Z}_n] = \frac{\sigma^2}{n} = \frac{p(1-p)}{n}.$$

and the Law of Large Numbers says as $n \rightarrow \infty$,

$$\bar{Z}_n \xrightarrow{P} \frac{\pi}{16}.$$

Therefore,

$$\boxed{\hat{\pi}_n = 16\bar{Z}_n \xrightarrow{P} \pi.}$$

This is a Monte Carlo estimator of π .

3.1 How Accurate is the Monte Carlo Estimate?

The Law of Large Numbers tells us that

$$\hat{\pi}_N \rightarrow \pi,$$

but the Central Limit Theorem tells us how the estimate fluctuates for large but finite N . Since

$$Z_i \sim \text{Bernoulli}\left(\frac{\pi}{16}\right),$$

we have

$$\mu = \frac{\pi}{16}, \quad \sigma^2 = \frac{\pi}{16} \left(1 - \frac{\pi}{16}\right).$$

Therefore, by the CLT,

$$\bar{Z}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) = \mathcal{N}\left(\frac{\pi}{16}, \frac{1}{n} \frac{\pi}{16} \left(1 - \frac{\pi}{16}\right)\right)$$

Since

$$\hat{\pi}_n = 16\bar{Z}_n,$$

we obtain

$$\hat{\pi}_N \approx \mathcal{N}\left(\pi, \frac{256}{n} \frac{\pi}{16} \left(1 - \frac{\pi}{16}\right)\right)$$

NOTE: In the above expression, I have used the identity $\text{Var}[cX] = c^2 \text{Var}[X]$. Hence,

$$\text{Var}(\hat{\pi}_n) = \frac{256}{n} \frac{\pi}{16} \left(1 - \frac{\pi}{16}\right); \quad \text{SD}(\hat{\pi}_n) = 16\sqrt{\frac{(\pi/16)(1 - \pi/16)}{n}}.$$

EXERCISE: How many samples n do you need to reduce the $\text{Var}(\hat{\pi}_n) \leq 10^{-4}$?

This illustrates the complementary roles of the two limit theorems:

LLN: $\hat{\pi}_n \rightarrow \pi$,

CLT: the error around π has an approximately Gaussian shape with width proportional to $n^{-1/2}$.

4 Estimating the Population Variance

So far we have assumed that the population has mean μ and variance

$$\sigma^2 = \mathbb{E}[(X - \mu)^2].$$

But in statistics, the population distribution is generally unknown. Given observations

$$X_1, \dots, X_n,$$

we estimate the unknown population mean using

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

A natural estimate of the population variance is therefore

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

This looks exactly like the population variance, except that the unknown population mean μ has been replaced by the sample mean $\bar{X}_n$. But is this estimator centered around the true variance σ^2 ? In other words, is

$$\mathbb{E}[\hat{\sigma}_n^2] = \sigma^2?$$

4.1 Expectation of the Sample Variance

Start with the identity

$$X_i - \bar{X}_n = (X_i - \mu) - (\bar{X}_n - \mu).$$

Squaring and summing,

$$\sum_{i=1}^n (X_i - \bar{X}_n)^2 = \sum_{i=1}^n [(X_i - \mu) - (\bar{X}_n - \mu)]^2.$$

Expanding,

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X}_n)^2 &= \sum_{i=1}^n (X_i - \mu)^2 \\ &\quad - 2(\bar{X}_n - \mu) \sum_{i=1}^n (X_i - \mu) \\ &\quad + n(\bar{X}_n - \mu)^2. \end{aligned}$$

It is easy to show that

$$\sum_{i=1}^n (X_i - \mu) = n(\bar{X}_n - \mu).$$

Therefore,

$$\sum_{i=1}^n (X_i - \bar{X}_n)^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X}_n - \mu)^2.$$

Now take expectations. For the first term,

$$\mathbb{E} \left[\sum_{i=1}^n (X_i - \mu)^2 \right] = n\sigma^2.$$

For the second term,

$$\mathbb{E}[(\bar{X}_n - \mu)^2] = \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

Hence,

$$\begin{aligned}\mathbb{E}\left[\sum_{i=1}^n (X_i - \bar{X}_n)^2\right] &= n\sigma^2 - n\frac{\sigma^2}{n} \\ &= (n-1)\sigma^2.\end{aligned}$$

Since

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2,$$

we obtain

$$\mathbb{E}[\hat{\sigma}_n^2] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2\right] = \frac{1}{n} \mathbb{E}\left[\sum_{i=1}^n (X_i - \bar{X}_n)^2\right] = \frac{n-1}{n} \sigma^2.$$

Therefore,

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

is a **biased estimator** of the population variance σ^2 . In particular,

$$\mathbb{E}[\hat{\sigma}_n^2] = \frac{n-1}{n} \sigma^2 < \sigma^2,$$

so $\hat{\sigma}_n^2$ systematically underestimates the true population variance.

NOTE: The bias can be substantial for small sample sizes, but becomes negligible as n increases. For example,

$$n = 4 \quad \implies \quad \mathbb{E}[\hat{\sigma}_n^2] = \frac{3}{4} \sigma^2 = 0.75 \sigma^2,$$

whereas

$$n = 100 \quad \implies \quad \mathbb{E}[\hat{\sigma}_n^2] = \frac{99}{100} \sigma^2 = 0.99 \sigma^2.$$

Thus, the relative downward bias is exactly $1/n$ and vanishes as $n \rightarrow \infty$.